FINAL LIVE STATUS

# AI Voice & Contact Center

## Phase 1-7 complete — what is LIVE vs CODE-READY-AWAITING-INFRA

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Build: PASS (5.3s, 101 static pages)

Tests: 188/188 pass (5 suites, 0 failures)

Schema: 59 → 95 Prisma models (36 new, additive)

Routes: 35+ new v1 API endpoints

UI: 5 new dashboard views (4,471 lines)

# Executive Summary

Phases 1-7 of the AI Voice & Contact Center product have been built end-to-end. The codebase is complete, all schema migrations are additive (no existing functionality removed), all 5 test suites pass (188 assertions), and the production build compiles cleanly. The dashboard now has 5 new views under a "Voice Center" sidebar section.

> Critical distinction: code-ready does NOT mean live. Every external service (FreeSWITCH, Piper, faster-whisper, BD SIP trunk) is wrapped in an adapter with a simulation fallback. When the operator provisions the real infrastructure, the adapters automatically use it. Until then, every call/TTS/ASR returns simulated=true with a clear warning — no false claims of real calls (per directive §2: "Keep demo functionality visibly separate from production functionality").

# LIVE vs CODE-READY Matrix

| # | Capability | Status | What makes it LIVE |
|---|---|---|---|
| 1 | MFA (TOTP) setup + verify | LIVE | Already works — no external dep |
| 2 | Refresh token rotation | LIVE | Already works — no external dep |
| 3 | Voice agent CRUD | LIVE | Already works — DB persistence |
| 4 | Consent capture + audit | LIVE | Already works — DB persistence |
| 5 | Phone number provisioning | LIVE (manual) | Provision via Twilio/Telnyx/BD carrier for real PSTN |
| 6 | Call state machine + events | LIVE | DB persistence; transitions fire events |
| 7 | Call originate (real) | CODE-READY | Install FreeSWITCH or set Twilio/Telnyx creds |
| 8 | Call recording (real) | CODE-READY | Same as #7 + consent flow already in place |
| 9 | IVR / ring groups / queues | CODE-READY | Models + CRUD ready; FreeSWITCH executes them |
| 10 | Bangla TTS (Piper) | CODE-READY | Install Piper + bn_BD-google-medium voice pack; set PIPER_BASE_URL |
| 11 | Bangla TTS (Z.AI fallback) | LIVE | Already configured — Z.AI cloud (free) works |
| 12 | Multilingual ASR | CODE-READY | Install faster-whisper or use Z.AI ASR (live) |
| 13 | Voice cloning (Tortoise) | CODE-READY | Install Tortoise + signed consent form flow |
| 14 | Outbound campaigns | LIVE (scheduling) | Launch validates calling hours + DNC; calls need #7 |
| 15 | Knowledge base (RAG) | LIVE (CRUD) | Document upload + chunking works; embedding needs Ollama bge-m3 |
| 16 | Agent workspace UI | LIVE | Dashboard view works; needs FreeSWITCH for real calls |
| 17 | Supervisor listen/whisper/barge-in | CODE-READY | Routes exist; needs FreeSWITCH conferencing |
| 18 | Customer 360 + shared inbox | LIVE | DB persistence; works end-to-end |
| 19 | Call analytics (sentiment/intent) | CODE-READY | Routes exist; needs AI call post-processing |
| 20 | Billing plans + subscriptions | LIVE | DB persistence; payment gateway is Phase 6+ |
| 21 | Usage metering | LIVE | Tracks calls/TTS/ASR per-tenant per-day; cost = $0 (free providers) |
| 22 | Credit balance + invoices | LIVE | DB persistence; payment provider replaceable |

# What's Already LIVE (no infra needed)

- MFA + refresh tokens — TOTP RFC 6238 (no external dep); 30-day refresh rotation
- Voice agent builder — full CRUD for AI agent configs (system prompt, greeting, tools, working hours, approval mode)
- Consent capture — every consent persisted with who/what/when/how/method/scope (per directive §10)
- Phone number provisioning — manual numbers can be added; real PSTN needs Twilio/Telnyx/BD carrier
- Call state machine — ringing → active → hold → transfer → park → closed; every transition logged as CallEvent
- Bangla TTS via Z.AI — already configured; falls back to simulation if rate-limited
- Outbound campaign scheduling — launch validates Asia/Dhaka calling hours + DNC + consent requirement
- Knowledge base CRUD — create KBs, upload documents, chunking infrastructure ready
- Agent workspace UI — status selector, call panel, dial pad, CRM sidebar, customer 360 timeline
- Customer 360 + shared inbox — unified inbox across voice/SMS/email/web_chat with AI-vs-provider-confirmed distinction
- Billing + usage metering — plans, subscriptions, credits, invoices, per-day usage rollups; $0 cost (free providers)
- 5 new dashboard views — Voice Agents, Calls, Voice Studio, Contact Center, Billing (4,471 lines)

# What's CODE-READY (needs infra to go LIVE)

| Capability | Infra needed | Effort |
|---|---|---|
| Real PSTN calls | FreeSWITCH (Docker) + SIP trunk (Twilio/Telnyx/BD carrier) | 1-2 days deploy + carrier partnership |
| Bangla TTS (Piper) | Install Piper + bn_BD voice pack on voice host | 4 hours; MIT license, CPU-only |
| Streaming ASR | Install faster-whisper (GPU recommended) or use Z.AI plug | 1 day; GPU optional |
| Voice cloning (Tortoise) | Install Tortoise + signed consent workflow | 2 days; Apache 2.0 license |
| Supervisor listen/whisper/barge-in | FreeSWITCH mod_conference + ESL events | 2 days; needs FreeSWITCH |
| Call analytics (sentiment/intent) | AI call post-processing worker | 3 days; uses existing LLM chain |
| Knowledge base RAG embeddings | Ollama + bge-m3 model | 4 hours; CPU-only |
| PostgreSQL RLS (multi-tenant hardening) | Migrate SQLite → PostgreSQL + RLS policies | 1 week; Phase 8 of master audit |
| BD PSTN (+880 numbers) | BTRC-licensed carrier partnership | Weeks; legal/regulatory |
| Bangla voice quality evaluation | Native speaker panel (3+ reviewers) | 2 weeks; per BANGLA_VOICE_MODEL_EVALUATION.md |

# Schema Statistics

| Layer | Before | After | Delta |
|---|---|---|---|
| Prisma models | 59 | 95 | +36 (additive only) |
| API routes (v0) | 150 | 150 | 0 (no changes) |
| API routes (v1) | 11 | 50+ | +35+ new voice/contact-center routes |
| Lib modules | 21 | 24 | +3 (telephony-adapter, voice-engine, asr-engine) |
| UI views | 17 | 22 | +5 (voice-agents, calls, voice-studio, contact-center, b |
| Test assertions | 127 | 188 | +61 (test-voice-contact-center.ts) |
| Destructive migrations | 0 | 0 | None — schema is additive only |

# Test Results — 188/188 PASS

| Suite | Assertions | Covers |
|---|---|---|
| verify-login | 4 | JWT issue + verify + forged token rejected |
| test-integrations | 36 | AES-256-GCM vault, PKCE, OAuth state, AI chain, Bengali greeting |
| test-ollama-cloud | 63 | Ollama Cloud, circuit breaker, PII filter, bounded agent loop, real LLM call |
| test-sms-providers | 24 | SMS provider chain, Twilio real call (401), Postiz adapter |
| test-voice-contact-center | 61 | Phase 1-6 schema + voice agent CRUD + consent + call state machine + usage metering + bi |
| Total | 188 | 0 failures |

# Production Build

✓ Running next.config.ts took 39ms
✓ Compiled successfully in 5.3s
✓ Generating static pages using 1 worker (101/101) in 259ms

Route (app) — 101 pages prerendered as static + dynamic API routes
No TypeScript errors. No build warnings. Exit code 0.

# Outstanding Blockers (operator actions)

- Install FreeSWITCH on a voice host with public IP + SIP trunk (Twilio/Telnyx for international test numbers; BTRC-licensed carrier for BD PSTN). Set FREESWITCH_ESL_HOST + PORT + PASSWORD env vars.
- Install Piper TTS + bn_BD-google-medium voice pack on the voice host. Set PIPER_BASE_URL. Verify Bangla quality per BANGLA_VOICE_MODEL_EVALUATION.md (100-sentence corpus + 3+ native speakers).
- Install faster-whisper (or use Z.AI cloud ASR already integrated). Set FASTER_WHISPER_URL for local; GPU recommended for streaming + low latency.
- (Optional) Install Tortoise TTS for authorized voice cloning. Set TORTOISE_BASE_URL. Voice clone authorization flow already implemented (POST /api/v1/voice-clone-authorizations).
- Migrate to PostgreSQL 16 for Row Level Security (Phase 8 of master audit). SQLite is currently staging-only.
- Source BD SIP trunk for real +880 numbers. Cloud providers cannot provision BD numbers directly — requires BTRC-licensed carrier partnership.
- Commission Bangla voice evaluation per BANGLA_VOICE_MODEL_EVALUATION.md (3+ native speakers, 100-sentence corpus, 4-dimension rubric, WER ≤15% threshold).
- Configure payment provider (Stripe / SSLCommerz for BD / manual invoicing) for billing to go fully live.
- Configure external cron (systemd timer / GitHub Actions) for /api/cron/social-publish + voicemail transcription queue.
- Production deployment approval — per directive §14 acceptance criteria: real inbound + outbound calls + Bangla TTS evaluation + tenant isolation tests + load testing must all pass before production.

# Bottom Line

Phase 1-7 codebase is complete and verified. All 7 phases have their code in place: foundation (MFA + RBAC + agent registry), telephony (FreeSWITCH adapter + call state machine), voice engine (Piper + faster-whisper adapters), AI call agents (orchestrator + campaigns + knowledge base), human contact center (agent workspace + supervisor + customer 360), billing (plans + usage + credits), and production readiness (188 tests pass + build clean).

> The product is code-ready. The remaining blockers are infrastructure provisioning (FreeSWITCH, Piper, faster-whisper, BD SIP trunk) and regulatory (BTRC license, Bangla voice evaluation, BD data protection compliance). All adapters are simulation-capable, so the operator can verify each capability as it goes live — no false claims of real calls until the real infrastructure is in place.